

ARYENDRA PANDEY

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SKILLS

Programming: Python, C++, JavaScript, SQL
Backend & Software: FastAPI, REST APIs, Git, Linux, CMake
AI / ML: PyTorch, YOLOv8, OpenCV, ONNX, INT8 Quantization, ChromaDB, Streamlit
Web Technologies: Next.js, TypeScript, React, Tailwind CSS
Tools & Platforms: GitHub, CUDA, Splunk, Wireshark, Verilog HDL

WORK EXPERIENCE

Software Developer Intern

April 2026 – May 2026

Tridit Digital

- Developed and customized AEM content components, templates, and publishing workflows for enterprise-scale web solutions.
- Integrated AI-driven features into AEM pipelines for automated content tagging and intelligent personalization, reducing manual editorial overhead.

IT Infrastructure Intern

June 2026 – Present

XtraNet Technologies Limited, Bhopal

- Administered and monitored enterprise network infrastructure, handling system configuration, fault diagnosis, and client-side troubleshooting.
- Managed server environment workflows supporting uptime and operational reliability of production systems.

PROJECTS

Brain Tumor Detection System with Hardware-Aware Inference Optimization

GitHub

PyTorch · YOLOv8n · ONNX · CUDA · INT8 Quantization · LayerCAM · Streamlit

- Trained YOLOv8n on 7,200 MRI scans across 4 tumor classes achieving **89.18% mAP@0.5** with CUDA-accelerated inference at 6.23 ms/image on an RTX 3050 GPU.
- Implemented FP32 → INT8 static quantization reducing model size by **47.89%** (5.94 MB → 3.09 MB) at only 0.48% mAP loss, validating edge deployment feasibility.
- Deployed LayerCAM explainability with IoU scoring and FP32 vs INT8 benchmarking in a real-time Streamlit diagnostic dashboard.

NeuroSentry — Real-Time Cyber Threat Detection System

GitHub

Python · FastAPI · ML Inference · Log Processing · React · Vite

- Built an end-to-end threat detection pipeline parsing system/audit logs, extracting behavioral features, and running online ML inference for real-time anomaly classification.
- Developed a modular FastAPI backend with REST endpoints for log ingestion, feature engineering, and live model inference; built a React + Vite dashboard visualizing live alerts and security statistics.

SoTrail — AI-Powered Travel Planning Platform

GitHub

FastAPI · ChromaDB · RAG · Llama · Amadeus API · Next.js · TypeScript

- Built a FastAPI backend with RAG architecture using ChromaDB and a Llama-powered chatbot for context-aware travel recommendations; integrated Amadeus API for real-time flight and hotel search.
- Designed a Next.js + TypeScript frontend with itinerary generator, chatbot panel, and analytics dashboard.

QoS Degradation Predictor — Network Systems ML

GitHub

C++17 · CMake · Logistic Regression · Network Simulation

- Designed a time-driven packet-switched network simulator in C++17 modeling multi-flow traffic with FIFO queues; implemented a logistic regression classifier for real-time QoS degradation detection, bridging systems simulation with applied ML.

EDUCATION

Manipal Institute of Technology, MAHE

2023 – Present

B.Tech — Electronics and Communication Engineering

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Object-Oriented Programming, Machine Learning

CERTIFICATIONS

Google Cybersecurity

AWS Cloud Practitioner

Meta Front-End Dev

Meta JavaScript

EXTRACURRICULAR ACTIVITIES

- Core Committee Member** — TechTatva 2025, Technical Fest, MIT Manipal
- Organising Committee Member** — Revels 2025, Cultural Fest, MIT Manipal
- Management Committee Member** — Aaina Dramatics, MIT Manipal